



# Dynamic Customer Segmentation and Predictive Evolution

Big Data Analytics Lab [CSE 3145]

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# Project Overview: Big Data Customer Analytics

Customer segmentation, while a basic marketing approach, gains significant complexity and opportunity when applied to big data. This project leverages the **Online Retail** dataset, which contains transactional data for a UK-based online retailer, offering a rich source for understanding customer purchasing behavior. The **RFM (Recency, Frequency, Monetary) model**, a well-established marketing analytics technique, is employed to segment customers based on their recent purchase activity, purchase frequency, and monetary value spent. By combining these traditional methods with contemporary big data technologies like PySpark, we aim to demonstrate a scalable and efficient approach to customer analytics.

## 1.1 Objectives

- To implement a robust data processing pipeline using PySpark for cleaning, transforming, and standardizing raw transactional data.
- To perform customer segmentation using the RFM model combined with K-Means clustering to identify distinct customer groups.
- To develop a predictive model using Gradient Boosted Trees (GBT) to forecast customer transitions between segments and anticipate future buying behavior.
- To demonstrate the scalability and efficiency of big data tools (HDFS, PySpark) in handling large-scale customer analytics.

# Dataset and Characteristics

We utilized the UCI Online Retail Dataset containing 541,909 UK e-commerce transactions from Dec 2010-Dec 2011 for comprehensive customer behavior analysis

## Methodology: Distributed Analytics Pipeline



### Data Ingestion & Exploration

Leverage HDFS for scalable storage and PySpark for initial data loading and exploratory analysis of raw transactional data.



### Data Preprocessing & RFM Feature Engineering

Implement PySpark transformations for cleaning, handling missing values, and calculating RFM (Recency, Frequency, Monetary) features from transactional data.



### Customer Segmentation (K-Means)

Apply the K-Means clustering algorithm on the RFM features using PySpark's MLlib for identifying distinct customer segments.



### Model Evaluation & Interpretation

Evaluate clustering quality and interpret the characteristics of each customer segment to derive actionable insights.



### Predictive Modeling (GBT)

Utilize Gradient Boosted Trees (GBT) in PySpark for forecasting future customer behavior, such as churn or segment transitions.

# Data Preprocessing & Feature Engineering

Initial dataset contained 541,909 transaction entries. After comprehensive cleaning, the data was refined to 397,884 confirmed transactions, ready for analysis.



## Remove NA Values

Eliminated entries with missing CustomerID values to ensure data integrity.



## Remove Cancelled Orders

Filtered out invoices with a 'C' prefix, indicating cancelled transactions.



## Convert InvoiceDate

Transformed raw InvoiceDate strings into a usable timestamp format.



## Remove Negative Values

Removed entries with non-positive Quantity or UnitPrice to correct erroneous data.



## Recency Calculation

Determined the recency of each customer's last purchase relative to a snapshot date.



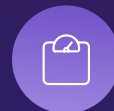
## Frequency Calculation

Calculated how often each customer makes purchases within the given period.



## Monetary Calculation

Aggregated the total monetary value spent by each customer.



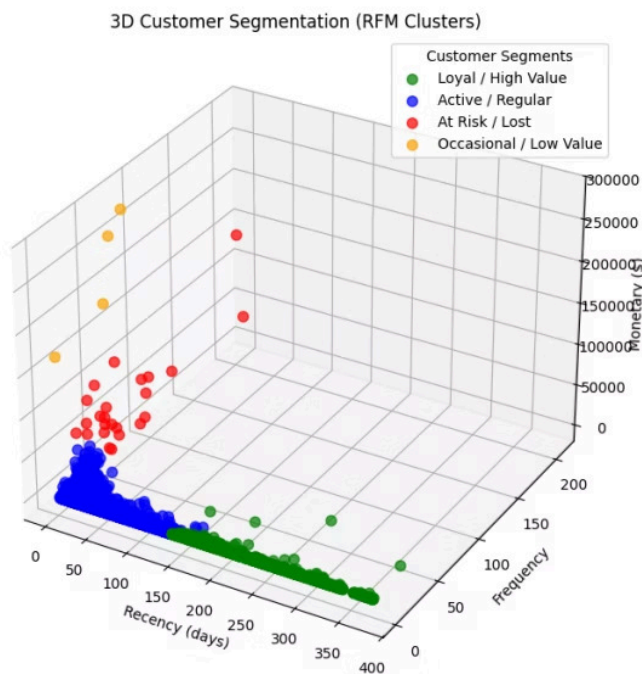
## Feature Standardization

Standardized numerical features using PySpark's StandardScaler for optimal model performance.



# Customer Segmentation Results (K=4)

The K-Means analysis identified four unique customer segments with distinct RFM patterns, forming the basis for data-driven customer management programs.



## Cluster 3: Elite Spenders

Avg. Monetary: \$22,576.4

Smallest, most valuable group (10 customers) with exceptional value and purchase frequency.

## Cluster 2: Loyal High-Value

Avg. Monetary: \$5,843.7

Consistent high spenders (228 customers) with sustained activity.

## Cluster 0: Active Regular

Avg. Monetary: \$1,945.8

Largest stable base (2,975 customers) with consistent engagement.

## Cluster 1: Inactive/Dormant

Avg. Monetary: \$715.4

Infrequent buyers (1,125 customers) with long recency intervals.

# Model Evaluation and Performance

Model testing confirmed the stability and predictive power of the combined K-Means and GBT approach, validating the scalability of the analytical design.

## K-Means Clustering

0.752

Silhouette Score

Indicates high-quality segmentation with strong inter-cluster separation.

The three RFM characteristics played a substantial role in segmentation, with monetary value having the highest discriminative effect.

## Gradient Boosted Trees (GBT)

| Metric            | Value        |
|-------------------|--------------|
| Accuracy          | 91.3%        |
| F1 Score          | 0.88         |
| Training Duration | 17.3 seconds |

The GBT model delivered strong predictive performance, accurately identifying potential customer switches between high-value and low-engagement clusters.

# Random Forest Classification Model

A Random Forest Classifier was implemented to predict customer segments based on RFM features. This model accurately classifies customers into predefined K-Means clusters.

**99.1%**

**Prediction Accuracy**

The Random Forest model achieved exceptional accuracy in classifying customers into their respective segments.

# Gradient Boosted Trees Regressor

A Gradient Boosted Trees Regressor model was utilized to predict the future monetary value of customers, leveraging their Recency and Frequency data.

## RMSE=7201

### Monetary Value Prediction

This metric indicates the average magnitude of the errors in predicting the customer's monetary value.

## NextSegment Categories & Thresholds

### >2000: Loyal/High Value

Customers with high predicted future spending.

### 501-2000: Active/Regular

Consistently engaged customers with moderate spending.

### 101-500: Occasional/Low Value

Infrequent purchasers or those with lower spending.

### ≤100: At Risk/Lost

Customers with very low predicted future monetary value.



# Strategic Business Implications

The robust distributed infrastructure, leveraging HDFS and PySpark, ensures efficient and fault-tolerant processing of vast transaction data. This technical foundation underpins the actionable insights derived from customer segmentation and predictive analytics.

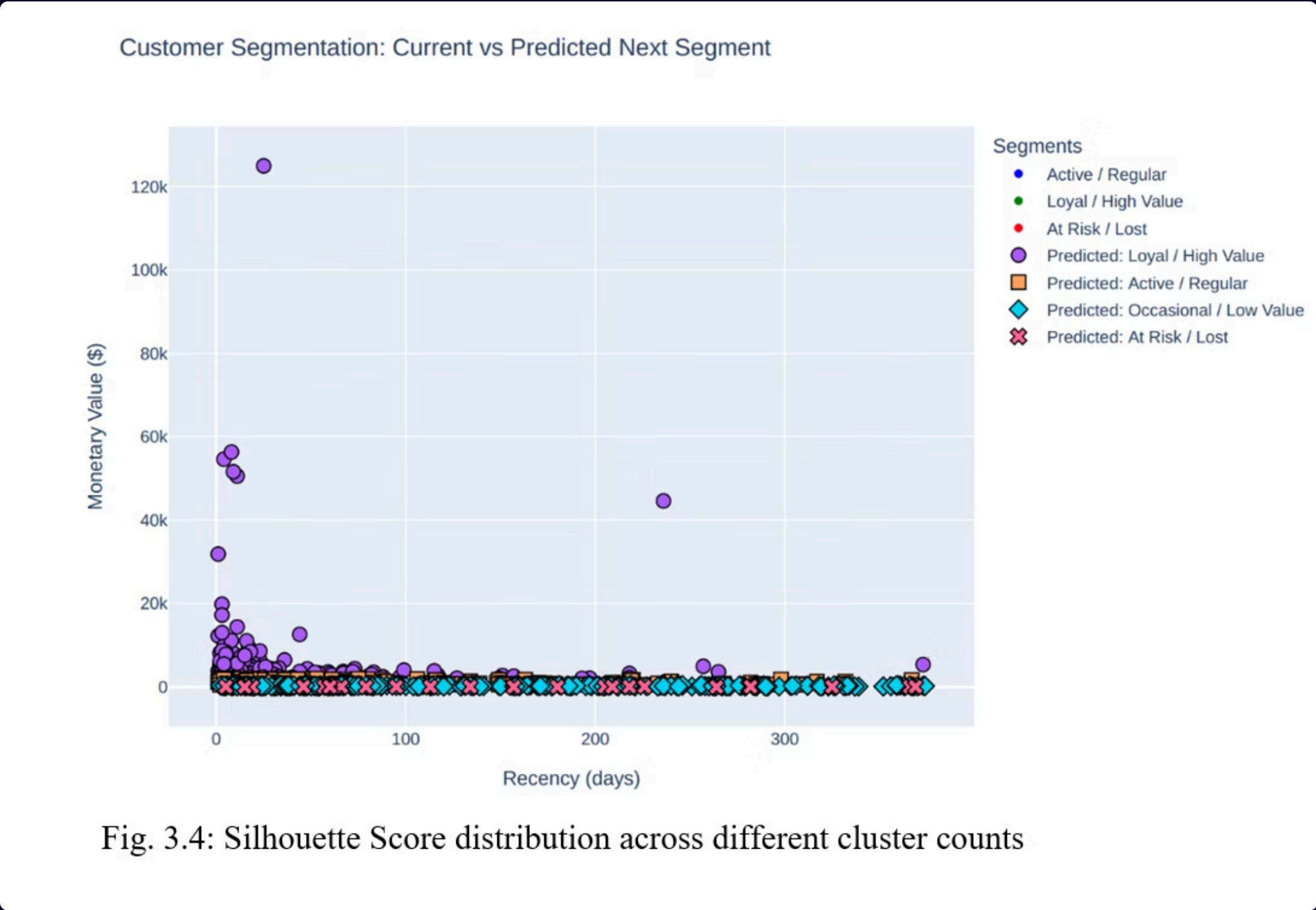
**Active Regular Customers**  
Engage through loyalty programs and personalized recommendations to maintain revenue flow. Focus on continuous retention efforts to secure this core segment.

**Inactive/Dormant Customers**  
Implement targeted re-engagement strategies, such as specific discount programs, to encourage renewed activity and enhance retention likelihood.

**Loyal High-Value Customers**  
Offer exclusive user privileges, premium service levels, and enhanced loyalty rewards to reinforce their value and strengthen brand allegiance.

**Elite Spenders**  
Prioritize these accounts with bespoke attention, focusing on delighting their loyalty and ensuring retention for future high-value purchases and services.

Predictive insights from the GBT model further empower organizations to proactively market and anticipate customer churn or transitions between clusters. Integrating these insights into CRM systems allows for the implementation of dynamic strategies that intentionally contribute to increased retention and revenue.





# Conclusion and Future Directions

This project successfully created a scalable, analytically strong framework for dynamic customer segmentation and predictive modeling within a big-data environment.

## Key Achievements

- Successfully integrated HDFS storage and PySpark computation for large-scale analytics.
- Achieved good cluster separation (Silhouette Score = 0.752) and high GBT predictive accuracy (91.3%).
- Bridged classical marketing analytics with modern distributed computing paradigms.

## Limitations & Future Work

- Expand to a full multi-node environment to assess full scalability under real-life enterprise workloads.
- Integrate Spark Streaming for real-time segmentation and continuous customer insights.
- Explore alternative clustering algorithms (GMM, DBSCAN) and ensemble predictive methods.